Module 3

Discrete-Time Dynamic Programming and Reinforcement Learning

In this document we present several MATLAB and Simulink homework problems relevant to dynamic programming and reinforcement learning in the discrete-time domain according to the material presented in Modules 1, 2, and 3.

MATLAB and Simulink Homework Problems

Homework Problem 3.1

Following Example 2.2 from Module 2, where the MATLAB live function for solving the liner-quadratic optimal control problem in *continuous-time* was considered, write a MATLAB code for a live function that solves the linear-quadratic optimal control problem in the *discrete-time* domain. Check controllability and observability by calling the corresponding live functions developed in Example 2.1 of Module 2. Note that the same controllability and observability tests apply to both continuous- and discrete-time domains.

Homework Problem 3.2

Write a code for a MATLAB live function that performs discretization of a linear system represented in the state space form with matrices A_c, B_c, C_c, D_c to obtain the corresponding discrete-time state space matrices A_d, B_d, C_d, D_d, for the given sampling period T_s. Using the obtained live function, demonstrate the code on the example of the continuous-time state space model of the islanded electric microgrid system whose state space matrices are given in Problem 2.9 of Module 2.

Homework **Problem 3.3**

Use the developed MATLAB live functions for examining system controllability and system observability developed in Example 2.1 of Module 2 to test controllability and observability of the discrete-time Boeing 747 aircraft model considered in Section 3.6 of Module 3.

Homework **Problem 3.4**

Build a Simulink block diagram to simulate dynamic behavior of the 14-th order computer communication network switch using the model data presented in Problem 2.4 from Module 2. Solve the corresponding LQ optimal discrete-time control problem with the quadratic performance penalty matrices being equal to identities.

Homework Problem 3.5

Write a code for a live function for examining stabilizability of a linear discrete-time dynamic system of order n represented by the state space matrices A and B. The system is stabilizable if the following conditions are satisfied for all unstable eigenvalues

$$\text{rank}\left[A-\lambda_i I \quad B\right]=n, \quad \forall i \quad \text{for which} \quad \left|\lambda_i(A)\right|>1, \quad i=1,2,...,n$$

In other words, the rank of the matrix $\left[A-\lambda_i I \quad B\right]=n$ is full for all unstable eigenvalue of matrix A. Demonstrate the live function on the example of the discretized model of an inverted pendulum on a cart from Section 3.5 of Module 3.

Homework Problem 3.6

Write a code for the live function that examines detectability of a linear discrete-time dynamic system of order n represented by the state space matrices A and C. The system is detectable if the following conditions are satisfied for all unstable eigenvalues

$$\text{rank}\begin{bmatrix} A-\lambda_i I \\ C \end{bmatrix}=n, \quad \forall i \quad \text{for which} \quad \left|\lambda_i(A)\right|>1, \quad i=1,2,...,n$$

In other words, the rank of the given matrix is full and equal to the system order for all unstable eigenvalue of matrix A.

Homework Problem 3.7

Write a MATLAB code for a live function that examines stability of a time-invariant discrete-time linear system. The live function should provide information about the number of unstable eigenvalues in an n-dimensional system. Demonstrate the live function on the example of an F-15 aircraft whose system matrix is given in Problem 2.11 in Module 2.